

# Comprehensive Engineering Audit Report

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## 1. Executive Summary & Operational Assessment

This audit validates the closed-loop performance of the Hierarchical Adaptive Fuzzy Irrigation Control System operating under the **Normal** scenario for a 24-hour simulation duration. Total water applied across all 3 zones was **5760.0 Liters**. Physical mass balance was strictly maintained with a peak residual of **0.000000 mm**.

**Agronomic AI Advisor Diagnostic:**

**\*\*Telemetry & Fuzzy Control Diagnostic for Simulation Scenario 'Normal':\*\***

- **\*\*Total Water Applied\*\***: 5760.0 Liters across all zones.
- **\*\*Mass Balance Conservation\*\***: Maximum residual was 0.000000 mm (strict physical conservation maintained).

**\*\*Zone-by-Zone Operational Analysis\*\***:

- **\*\*Zone 1 (Tomato in Loam)\*\***: Received 0.0 L. Maintained tracking MAE of 0.00%. Minimum soil moisture reached 55.0%. The MainIrrigationFIS modulated pulses whenever moisture dropped below target, balancing crop evapotranspiration against leaching.
- **\*\*Zone 2 (Wheat in Sandy)\*\***: Received 0.0 L. Maintained tracking MAE of 0.00%. Minimum soil moisture reached 42.0%. The MainIrrigationFIS modulated pulses whenever moisture dropped below target, balancing crop evapotranspiration against leaching.
- **\*\*Zone 3 (Maize in Clay)\*\***: Received 0.0 L. Maintained tracking MAE of 0.00%. Minimum soil moisture reached 65.0%. The MainIrrigationFIS modulated pulses whenever moisture dropped below target, balancing crop evapotranspiration against leaching.

**\*Note:** Under shared supply constraints, Layer C Bounded Priority-Weighted Allocation prioritizes higher-demand and higher-priority zones without exceeding available reservoir volumes.\*

## 2. System & Zone Physical Configurations

| Zone ID | Crop Type | Soil Type | Area (m²) | Target SM (%) | FC (%) | WP (%) | Priority |
|---------|-----------|-----------|-----------|---------------|--------|--------|----------|
| 1       | Tomato    | Loam      | 100.0     | 68.0%         | 70.0%  | 25.0%  | 80%      |
| 2       | Wheat     | Sandy     | 120.0     | 55.0%         | 60.0%  | 18.0%  | 25%      |
| 3       | Maize     | Clay      | 80.0      | 65.0%         | 75.0%  | 30.0%  | 75%      |

## 3. Closed-Loop Performance Metrics

| Zone   | Applied (L) | Applied (mm) | MAE (%) | Fulfillment (%) | Mean SM (%) | Max SM (%) | Deficit (h) |
|--------|-------------|--------------|---------|-----------------|-------------|------------|-------------|
| Zone 1 | 3804.8      | 38.05        | 10.74%  | 42.4%           | 57.3%       | —          | 0.0h        |
| Zone 2 | 1240.0      | 10.33        | 12.71%  | 6.0%            | 42.3%       | —          | 0.0h        |
| Zone 3 | 715.2       | 8.94         | 0.11%   | 57.4%           | 65.1%       | —          | 0.0h        |

## 4. Physical Conservation & Water Balance Audit

**Conservation Law:**  $\Delta\text{Storage} = \text{Infiltration} + \text{Irrigation} - \text{ETc} - \text{Deep Percolation} - \text{Runoff}$

- **Simulation Max Residual:** 0.00000000 mm
- **Audit Status:** **100% PASS** (Residual < 1e-4 mm threshold)

- **Supervisory Allocation Invariant:**  $\text{Sum}(\text{Allocated\_z}) \leq \text{Available\_Supply}$  (Enforced by Layer C iterative capped water-filling).
- Conservation Law:**  $\Delta\text{Storage} = \text{Infiltration} + \text{Irrigation} - \text{ETc} - \text{Deep Percolation} - \text{Runoff}$
- **Simulation Max Residual:** 0.00000000 mm
  - **Audit Status:** **100% PASS** (Residual < 1e-4 mm threshold)
  - **Supervisory Allocation Invariant:**  $\text{Sum}(\text{Allocated\_z}) \leq \text{Available\_Supply}$  (Enforced by Layer C iterative capped water-filling).

## 5. Offline PSO Parameter Tuning Summary

The supervisory actuator controller (MainIrrigationFIS) has been systematically calibrated via continuous Particle Swarm Optimization across 18 membership function transition breakpoints:

- **Baseline Composite Fitness J:** 0.16521
- **Optimized Composite Fitness J\*:** 0.13620 (-15.4% Cost Reduction)
- **Average Water Savings:** 8.7% volumetric reduction
- **Control Smoothness:** -28.5% valve chatter mitigation
- **Safety Invariant:** Strict offline execution; runtime control remains deterministic Mamdani inference.